Main Body

AgBioData Annual Meeting 2026: Toward Sustainable Agricultural Genomics Infrastructure

Running title: Sustainable agricultural genomics infrastructure

Authors

Olivia C. Haley - https://orcid.org/0000-0001-7237-8895

Marcela K. Tello-Ruiz - https://orcid.org/0000-0002-7499-5368

Monica Poelchau - https://orcid.org/0000-0002-4584-6056

Sunita Kumari - https://orcid.org/0000-0002-6605-6985

Sarah Dyer - https://orcid.org/0000-0001-5690-9633

Doreen Ware - https://orcid.org/0000-0002-8125-3821

David Molik - https://orcid.org/0000-0003-3192-6538

Keywords:

FAIR data; AI-ready data; biocuration; database sustainability; federated data infrastructure

Abstract

Agricultural genomics increasingly depends on distributed data resources that integrate genomes, transcriptomes, variation data, phenotypes, and associated metadata. Persistent challenges, including inconsistent standards, limited interoperability, incomplete metadata, and unstable funding, continue to constrain data reuse and downstream applications such as machine learning. This meeting report summarizes discussions and outcomes from the 2026 AgBioData Annual Meeting, which brought together members of the agricultural breeding, genetics, and genomics database community to identify practical strategies for developing sustainable agricultural genomics infrastructure that can support artificial intelligence. Building on prior AgBioData Consortium recommendations, we synthesize meeting themes and situate them within the broader literature on findable, accessible, interoperable, and reusable data; readiness for artificial intelligence; data stewardship; and database sustainability.

Three core transitions emerged from the meeting: implementing findable, accessible, interoperable, and reusable practices during data generation and publication rather than through retrospective curation; extending those stewardship practices with the

additional structure, provenance, and access needed for computational and artificial-intelligence-driven reuse, without equating broad scientific reuse with optimization for artificial intelligence; and developing federated, sustainably supported infrastructure that preserves community expertise while enabling interoperability across resources. Across these transitions, participants identified observability and attribution, socio-technical capacity, and international coordination as cross-cutting priorities. Together, these outcomes provide an actionable framework for designing and evaluating next-generation agricultural genomics resources.

1. Key Meeting Outcomes

Drawing on meeting discussions and the relevant literature, this report identifies six recommendations for agricultural genomics databases, journals, repositories, funders, and research communities:

- Embed FAIR (i.e., Findable, Accessible, Interoperable, and Reusable) practices at publication. Journals, reviewers, repositories, and community databases should coordinate before publication is complete to validate domain-appropriate metadata, persistent identifiers, ontology terms, repository selection, licensing, and links between publications and datasets.
- Consider AI-readiness through the lens of FAIR data stewardship. Agricultural genomics data intended for computational reuse should be structured, versioned, provenance-aware, programmatically accessible, and documented well enough to support reproducible machine-learning workflows.
- Build observability and attribution into data infrastructure. Dataset-level identifiers, versioning, API usage metrics, provenance records, and machine-readable citation practices can help document reuse, support reproducibility, and make data-stewardship labor more visible.
- Prioritize federation over full centralization. Agricultural genomics databases should retain community-specific expertise while exposing shared standards, resolvable identifiers, APIs, metadata schemas, and exchange formats that enable cross-resource discovery and reuse.
- Fund agricultural databases as long-term infrastructure. Sustainable support should include not only new data generation and tool development, but also maintenance, curation, and documentation, as well as workforce continuity, governance, user support, and shared services.
- Broaden international partnerships and strengthen global coordination. AgBioData should deepen its coordinating role in the agricultural genomics database ecosystem by growing international participation; building partnerships with international research organizations, data infrastructures, and standards communities; engaging industry; and pursuing bilateral and multilateral funding opportunities.

2. Introduction

AgBioData (https://www.agbiodata.org/) is a consortium of agricultural Breeding, Genetics, and Genomics (BGG) databases and other allied resources that works to improve data sharing, interoperability, biocuration, and long-term sustainability across the agricultural biological sciences. The consortium brings together database teams and their user communities (stakeholders) that support crop, livestock, model organism, and related community resources, with the goal of coordinating standards and practices for reusable agricultural data (Harper et al., 2018; Clarke et al., 2023). Building on this community role, the 2026 AgBioData Annual Meeting focused on how agricultural databases can move from shared recommendations toward implementable infrastructure for FAIR, AI-ready, and sustainable data reuse.

Discussions at the annual meeting built upon a series of community workshops held during the Plant and Animal Genome (PAG) Conference earlier that year. These workshops brought together database developers, curators, researchers, and infrastructure specialists to identify emerging challenges in agricultural genomics, including AI-ready data stewardship, database sustainability, interoperability, and FAIR implementation. Outcomes from the PAG workshops informed the agenda and working-group priorities of the annual meeting, while more detailed summaries of the workshop discussions are available through AgBioData community reports and the AgBioData website.

The meeting recognized that agricultural genomics now depends on an ecosystem of distributed databases that curate, integrate, and disseminate genome assemblies, gene annotations, variation datasets, germplasm records, phenotypes, environmental metadata, trait ontologies, and breeding-relevant analytical outputs (Harper et al., 2018; Clarke et al., 2023; Deng et al., 2023). These resources are not peripheral to research; they increasingly function as the computational substrate on which comparative genomics, genotype-phenotype association, genomic prediction, crop improvement, and artificial intelligence (AI)-enabled discovery depend (Deng et al., 2023; Hafner et al., 2025). As a result, the quality, interoperability, governance, and sustainability of agricultural databases directly shape what kinds of biological questions can be asked and answered.

Despite major progress, the agricultural genomics data ecosystem remains fragmented. Persistent challenges include inconsistent metadata, limited ontology alignment, heterogeneous data submission practices, insufficient links between publications and databases, and unstable funding for long-term curation (Harper et al., 2018; Clarke et al., 2023; Hafner et al., 2025). Prior AgBioData recommendations emphasized the importance of FAIR principles, biocuration, metadata standards, and sustainable support for genomics and genetics databases (Harper et al., 2018; Tello-Ruiz et al.,

2026). Additional work has shown that these issues remain active barriers to data reuse, particularly for genotype–phenotype integration and machine-learning applications (Andrés-Hernández et al., 2021; Deng et al., 2023; Hafner et al., 2025).

3. Meeting Scope and Participants

The 2026 AgBioData Annual Meeting marked a shift from identifying broad community challenges to defining implementable infrastructure solutions. Several converging developments, including growing attention to FAIR implementation, increasing use of AI methods, growing concern over database sustainability, and new community standards for interoperability, made 2026 an appropriate time to revisit the role of agricultural genomics databases as long-term research infrastructure. The 49 participants (18 in person, 31 virtual) included librarians, heads of breeding, genetics and genomics (BGG) databases, research scientists, software developers, and academic publishers. The hybrid meeting was held April 20–22 in Olathe, Kansas, USA, and was structured as four outcome-driven half-day sessions combining community discussion with working sessions. These sessions focused on grant objectives and progress, working group goal-setting, database sustainability, and network sustainability. The meeting schedule is provided in Supplementary File 1. Accordingly, the annual meeting was organized to translate the priorities identified during the PAG workshops into coordinated working-group activities and actionable community recommendations.

The authors synthesized meeting transcripts, automated session summaries, meeting notes, and working-group reports, and then situated the recurring themes within the relevant literature. Detailed summaries of the working-group discussions are provided in Supplementary File 2.

Building on the FAIR principles for scientific data stewardship (Wilkinson et al., 2016), the discussions emphasized three structural transitions: embedding FAIR practices within data-generation and publication workflows, extending FAIR stewardship to support AI-ready and machine-actionable data, and developing federated and sustainable infrastructure for agricultural genomics. The recommendations presented here reflect discussions among participating AgBioData members and collaborators and should be understood as community priorities rather than formal standards.

4. From FAIR Compliance to FAIR-at-Publication

4.1 Limitations of post hoc curation

A persistent issue in agricultural genomics is that data associated with publications are frequently not truly reusable, even when nominally “open.” The central problem is not

simply whether data are deposited and available, but whether they retain enough biological, experimental, computational, and provenance context to support reuse by researchers and databases. Datasets may be available as supplementary files, spreadsheets, or repository records while still lacking standardized metadata, controlled vocabularies, accession links, persistent identifiers, genome assembly versions, software versions, or machine-readable relationships among genotype, phenotype, environmental, and analytical data. Metadata availability, or the lack thereof, also remains a critical issue in genomics publication practices (Huang et al., 2025).

This reuse problem is especially acute in agricultural genomics because such datasets are rarely interpretable in isolation. Genomic datasets, (e.g., genotype matrices, genome assemblies, GWAS results, QTL tables, pan-genome annotations, and phenotypic datasets) often depend on associated information about germplasm identity, trait measurement protocols, environmental conditions, marker coordinates, genome build, experimental design, statistical models, and quality-control procedures. Studies of agricultural genetics, genomics, and breeding databases have repeatedly identified data standards, metadata completeness, ontology use, and interoperability as central barriers to data integration and reuse (Harper et al., 2018; Pommier et al., 2019; Andrés-Hernández et al., 2021; Clarke et al., 2023; Deng et al., 2023; Hafner et al., 2025; Huang et al., 2025).

FAIR principles provide a conceptual framework for addressing these challenges, but their value depends on early implementation (Wilkinson et al., 2016; Wise et al., 2019). When FAIR practices are applied only after publication, key contextual information may already be missing, inconsistent, or embedded in narrative text, figure legends, local file names, laboratory notebooks, or undocumented scripts. This lack of consistency is inefficient and the metadata often incomplete, because the tacit knowledge needed to interpret the data may be lost when projects end or personnel move on. At that point, curation becomes a process of reconstruction rather than validation. More broadly, work on FAIR implementation and reproducible computational research emphasizes that metadata are not merely descriptive supplements; they are part of the research object needed to support interpretation, reuse, and reproducibility (Leipzig et al., 2021; Wilson et al., 2021).

Post hoc curation also shifts the labor burden from authors and publishers to community databases. Agricultural genomics databases must often interpret heterogeneous supplementary files, reconcile inconsistent sample and germplasm identifiers, map traits to ontologies, validate genome build information, and convert idiosyncratic tables into reusable formats. This burden is difficult to scale and creates inequities among resources: well-funded databases may be able to rescue poorly structured datasets, whereas smaller community databases may lack the personnel and infrastructure to do

so. The AgBioData consortium has previously emphasized that sustainable agricultural genomics resources depend on biocuration, metadata standards, ontology alignment, and long-term database support, yet these activities remain under-resourced relative to their scientific importance (Harper et al., 2018; Clarke et al., 2023).

Delayed curation further reduces reproducibility and downstream integration. If datasets are deposited without clear provenance, versioning, identifiers, or links to analytical workflows, later users may be unable to determine which genome assembly, trait protocol, germplasm accession, software version, or quality-control procedure was used. These gaps are particularly consequential for comparative genomics, genotype–phenotype integration, genomic prediction, pan-genomics, and machine-learning applications, where small inconsistencies in identifiers, coordinate systems, or metadata can propagate into downstream analyses (Deng et al., 2023; Hafner et al., 2025). Thus, post hoc curation is not merely inefficient; it limits the scientific value of published data and constrains the transition from open data toward reusable, interoperable, and AI-ready agricultural genomics infrastructure.

AgBioData has devoted sustained attention to the conditions necessary for agricultural genomics data to be reused effectively. This work has progressed from consortium-wide recommendations concerning biocuration, metadata, ontologies, programmatic access, and database sustainability (Harper et al., 2018), through assessments of data-sharing and ontology practices and analyses of persistent barriers to reuse (Clarke et al., 2023; Hafner et al., 2025), to implementable standards for genomic file formats, reference-genome provenance, assembly and gene nomenclature, and persistent identifiers for genetic variants (Saha et al., 2022; Wright et al., 2024; Cannon et al., 2025; Tello-Ruiz et al., 2026). Collectively, these efforts treat reuse not as a final act of data deposit, but as an outcome that must be designed into data formats, metadata, identifiers, publication practices, and database infrastructure from the outset.

4.2 Embedding FAIR at publication

The meeting emphasized that FAIR implementation should be embedded within publication workflows rather than applied retrospectively. Participants recognized that this shift would require coordinated changes in the roles of journals, repositories, and community databases. Instead of treating data deposition as a final compliance step, publication could serve as a checkpoint at which metadata completeness, identifier consistency, repository selection, licensing, and links to domain databases are validated. Various approaches to this end goal have been suggested or implemented. The Publish-Review-Curate model proposes synchronized work among authors, peer reviewers, data experts, and journal editors during publication (Jung and Ahn, 2025). More operational examples include omics data-paper workflows that import standard-compliant metadata from repositories such as ENA, ArrayExpress, and

BioSamples into manuscript templates, thereby linking metadata enhancement, peer review, data auditing, and publication (Dimitrova et al., 2021). Broader data-publishing models have likewise treated publication as a coordinated workflow involving repository selection, metadata, persistent identifiers, licensing, data citation, and links between datasets and articles (Austin et al., 2017; Dallmeier-Tiessen et al., 2017). Reproducible publication workflows extend this logic by incorporating artefact review and metadata creation between manuscript submission and final publication (Peer et al., 2022). Domain infrastructures such as PLANTdataHUB and BD2K-LINCS further demonstrate that automated validation, continuous quality control, metadata standardization, and persistent identifiers can be embedded into production-scale data publication and curation environments (Stathias et al., 2018; Weil et al., 2023). These examples show that FAIR-at-publication is not merely aspirational: versions of this approach have already been implemented in data journals, institutional and disciplinary repositories, and domain-specific data coordination centers, suggesting that agricultural genomics venues could adopt similar editorial checks through phased, community-informed policy changes.

For agricultural genomics, FAIR-at-publication would require more than a generic data availability statement. Authors should be guided toward domain-appropriate repositories and asked to provide structured metadata for germplasm, traits, environments, genome assembly versions, software workflows, and analytical outputs (Andrés-Hernández et al., 2021; Deng et al., 2023; Hafner et al., 2025). Journal workflows could also integrate automated checks for accession numbers, persistent identifiers, ontology terms, file formats, and repository links. Such checks would reduce the downstream burden on curators and increase the likelihood that published datasets can be reused by both humans and machines (Leipzig et al., 2021; Wilson et al., 2021). This model reframes journals as entry points into data infrastructure rather than endpoints of dissemination. It also creates an opportunity for community-owned and community-led journals to help define community expectations for reusable genomics data, especially by recommending discipline-specific repositories, minimum metadata standards, and machine-readable data availability statements (Wilkinson et al., 2016; Harper et al., 2018; Rocca-Serra et al., 2023).

5. AI-Readiness in Relation to FAIR Data Stewardship

5.1 FAIR is necessary but insufficient for AI readiness

FAIR data are not automatically AI-ready, or vice-versa. FAIR principles improve the conditions for data discovery, access, interoperability, and reuse, but machine-learning workflows impose additional requirements (Wilkinson et al., 2016; Leipzig et al., 2021). AI-ready datasets must be sufficiently structured, standardized, labeled, versioned, and documented to support computational training, evaluation, and reproducible reuse. In

agricultural genomics, this distinction is critical because many datasets are findable and accessible but remain difficult to use for modeling due to inconsistent trait definitions, missing environmental metadata, unclear sample identifiers, heterogeneous file formats, or absent quality-control information (Andrés-Hernández et al., 2021; Deng et al., 2023; Hafner et al., 2025).

For example, a phenotypic dataset may be accessible but not AI-ready if trait measurements are not linked to ontology terms, if trial environments are incompletely described, or if genotype identifiers cannot be reconciled with germplasm records (Andrés-Hernández et al., 2021; Clarke et al., 2023). Similarly, a genome annotation dataset may be available but difficult to use in downstream models if assembly versions, annotation pipelines, evidence codes, or provenance metadata are unclear (Deng et al., 2023; Hafner et al., 2025). These issues affect not only model performance but also interpretability, reproducibility, and fairness of downstream predictions (Leipzig et al., 2021; Wilson et al., 2021).

FAIR provides many of the foundational data-stewardship practices that also support AI-ready data, while AI-ready datasets may require additional technical, semantic, and governance considerations beyond the FAIR principles. There is a subtle but important distinction here: FAIR enables many characteristics associated with AI-ready data, but FAIR should not become synonymous with AI optimization. The purpose of FAIR remains broad scientific reuse and interoperability. AI is an increasingly important, but not inherently values-driven, beneficiary of those practices; AI does not, and cannot, define the values that guide data stewardship.

5.2 Defining AI-ready data infrastructure

The AgBioData Consortium has already recognized the monumental challenge of AI-readiness at agricultural genomics databases of varying scope and financial resources, having established the AI-Ready Working Group (AIR WG) in early 2026. The AgBioData Annual Meeting identified AI-readiness as an infrastructural property rather than a feature of individual datasets alone. A dataset becomes AI-ready when it is embedded in a system that supports standardized description, reliable access, provenance tracking, quality control, licensing clarity, and computational reuse (Leipzig et al., 2021; Rocca-Serra et al., 2023). This is especially important in agricultural genomics, where AI applications often require integration across multiple resources, including genome annotations, marker datasets, phenotypes, environmental records, germplasm repositories, and literature-derived knowledge (Deng et al., 2023; Hafner et al., 2025).

AI-ready agricultural genomics infrastructure should therefore include five core components. First, metadata should be structured and linked to community vocabularies or ontologies (Andrés-Hernández et al., 2021; Clarke et al., 2023). Second, provenance

should document the origin, transformation, quality-control procedures, and version history of data objects (Leipzig et al., 2021; Wilson et al., 2021). Third, access should be scalable through APIs (Application Programming Interfaces), bulk downloads, and machine-readable formats (Rocca-Serra et al., 2023). Fourth, interoperability should allow identifiers, traits, germplasm records, genome coordinates, and environmental variables to be resolved across databases (Clarke et al., 2023; Deng et al., 2023). Fifth, governance should clarify licensing, attribution, responsible reuse, and long-term stewardship (Harper et al., 2018; Lin et al., 2024).

This definition makes AI-readiness testable. Resources can be evaluated by whether their data are programmatically accessible, whether metadata are complete and ontology-linked, whether data versions are preserved, whether identifiers resolve across systems, and whether downstream users can cite or credit datasets accurately (Wilkinson et al., 2016; Leipzig et al., 2021; Haley et al., 2026).

5.3 Observability and attribution

A major gap in current agricultural genomics infrastructure is limited observability into how datasets are reused after publication or database release. Data may be downloaded, transformed, incorporated into machine-learning pipelines, or used to train models without clear mechanisms for tracking use, attribution, or downstream impact (Leipzig et al., 2021; Wilson et al., 2021). This creates practical and ethical challenges. Data generators may not receive credit, curators may be unable to demonstrate the value of their work, and downstream users may be unable to reconstruct which versions of datasets contributed to a model or result (Harper et al., 2018; Lin et al., 2024).

In this context, observability refers to the ability of data infrastructure to record, expose, and analyze how data objects are accessed, transformed, cited, and reused. Possible mechanisms include persistent identifiers for datasets and versions, machine-readable citation metadata, API access logs, provenance graphs, dataset-level usage metrics, and standardized attribution practices (Wilkinson et al., 2016; Leipzig et al., 2021; Lin et al., 2024). These mechanisms would help connect data reuse to scholarly credit and infrastructure evaluation.

For AI applications, observability is also a reproducibility requirement. If models are trained on changing or poorly versioned datasets, later users may be unable to determine which data supported a given result (Leipzig et al., 2021; Haley et al., 2026). Agricultural genomics resources therefore need provenance and attribution systems that can operate across distributed databases, publications, and computational workflows (Harper et al., 2018; Clarke et al., 2023; Hafner et al., 2025).

6. Federated Infrastructure and Interoperability

6.1 Limits of centralization

The meeting recognized that full centralization of agricultural genomics data is neither practical nor desirable. Agricultural data span many species, communities, institutions, data types, and governance contexts. Crop-specific databases often contain specialized knowledge about nomenclature, germplasm, traits, genome assemblies, community standards, and user needs that cannot be easily transferred to a single central repository (Harper et al., 2018; Clarke et al., 2023). At the same time, users increasingly need to ask questions across resources, species, and data types (Deng et al., 2023; Hafner et al., 2025).

Federated infrastructure offers a way to balance local stewardship with cross-resource integration. In a federated model, databases retain authority over their domain-specific data while exposing standardized metadata, identifiers, APIs, and exchange formats that allow discovery and integration across systems (Clarke et al., 2023; Deng et al., 2023). This approach is well suited to agricultural genomics because it preserves community expertise while enabling broader interoperability.

A key implication is that the goal should not be to collapse agricultural genomics resources into a single database. Rather, the goal should be to make distributed resources function as a coordinated infrastructure layer through shared standards, resolvable identifiers, common APIs, and governance agreements (Harper et al., 2018; Clarke et al., 2023; Hafner et al., 2025).

6.2 Shared services and data visitation

The meeting highlighted shared services as a practical route toward federation. Rather than requiring each database to independently build every component, the community could coordinate around reusable infrastructure such as metadata validators, ontology lookup services, identifier resolution systems, genome browsers, annotation pipelines, cross-database APIs, and repository recommendation tools. Shared services would reduce duplication and allow smaller databases to adopt community standards without rebuilding infrastructure from scratch (Harper et al., 2018; Clarke et al., 2023; Rocca-Serra et al., 2023).

Data visitation represents a complementary model. Instead of moving all data into a single, shared repository, computational workflows can be brought to the data through standardized APIs, containerized analyses, workflow systems, or federated query layers. This approach is particularly relevant for large genomic datasets, distributed institutional resources, and cases where governance constraints limit data movement (Deng et al., 2023; Lin et al., 2024). For agricultural genomics, data visitation could

support cross-resource analyses while preserving local stewardship and reducing the cost of full centralization.

The practical challenge is governance: shared services require agreements about maintenance, versioning, authentication, documentation, service-level expectations, and funding. Without these commitments, shared tools risk becoming short-lived prototypes rather than durable infrastructure (Harper et al., 2018; Lin et al., 2024).

7. Sustainability and Funding Models

7.1 Structural funding gaps

A recurring theme of the meeting was the mismatch between how agricultural databases are used and how they are funded. These resources function as long-term scientific infrastructure, but they are often supported through short-term, project-based grants (Harper et al., 2018; Imker, 2025). Such funding mechanisms are poorly aligned with the ongoing work required to maintain database reliability, update genome assemblies, curate annotations, support users, preserve identifiers, modernize software, and integrate new data types (Harper et al., 2018; Lin et al., 2024).

This mismatch produces several structural risks. First, databases may lose expert curators and developers when grant cycles end. Second, resources accumulate technical debt when maintenance and modernization are not funded. Third, valuable datasets may become inaccessible or poorly documented if databases cannot sustain curation. Fourth, community trust declines when resources become unstable, outdated, or unavailable (Harper et al., 2018; Hafner et al., 2025). These risks are especially consequential in agricultural genomics because many databases serve as reference infrastructure for entire crop or organism communities (Clarke et al., 2023; Deng et al., 2023).

Sustainability should therefore be evaluated not only in financial terms but also in terms of workforce continuity, governance, technical maintainability, and community value. A database that cannot retain expertise or maintain core services cannot fully support FAIR or AI-ready data reuse (Harper et al., 2018; Lin et al., 2024; Hafner et al., 2025).

7.2 Portfolio funding strategies

The meeting proposed portfolio funding as a more realistic sustainability strategy than reliance on a single grant mechanism. Agricultural genomics infrastructure serves multiple functions: it supports basic research, applied breeding, education, data stewardship, community coordination, software development, and AI-enabled discovery (Harper et al., 2018; Deng et al., 2023). No single funding stream is likely to support all of these functions indefinitely.

A portfolio model could distribute support across complementary sources. Federal infrastructure programs could support core cyberinfrastructure, long-term maintenance, and shared services. Domain-specific agencies could support crop- or trait-specific data integration. Philanthropic or foundation support could fund coordination, training, and community engagement. Institutional commitments could support personnel continuity and hosting. Partnerships with journals or repositories could support publication-linked data validation and deposition workflows (Harper et al., 2018; Lin et al., 2024).

The key principle is that funding should match infrastructure function. Short-term research grants may be appropriate for innovation and prototyping, but stable infrastructure requires mechanisms that support maintenance, governance, documentation, user support, and workforce retention. Without such mechanisms, databases remain vulnerable even when they are scientifically essential (Harper et al., 2018; Clarke et al., 2023; Hafner et al., 2025; Imker, 2025).

8. International Partnerships and Global Coordination

International coordination is an implementation requirement for federated agricultural genomics infrastructure, not a separate outreach activity. Standards, identifiers, metadata, and computational services must operate across national and institutional boundaries if distributed resources are to function as a coherent network. A dedicated meeting session therefore examined how AgBioData could broaden international participation and strengthen its coordinating role through data integration, standards development, shared AI modeling, and federated data networks.

8.1 Gaps, opportunities, and community strengths

The session focused on identifying key gaps and opportunities for AgBioData in the international arena over the next three years. In a participant poll, funding and data integration were ranked as the top challenges, with funding receiving the most votes. Discussion then turned to opportunities for international collaboration, highlighting AgBioData's strength in providing infrastructure for coordination and communication and emphasizing the value of the community's shared knowledge. Participants also identified opportunities for shared resources and platforms that could reduce costs and improve collaboration across the international community.

8.2 Global data integration and underrepresented regions

Opportunities for international collaboration centered on data integration and AI model coordination. Participants identified the need for a unified metadata system to help harmonize data across databases globally, an objective that aligns directly with the federated infrastructure model described in Section 6. The discussion highlighted underrepresented regions, particularly Asia and Africa, and potential partnerships with

CGIAR, a global agricultural research partnership; the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT); and the Food and Agriculture Organization of the United Nations (FAO). Participants also considered collaborating with crop-focused research institutions, such as the International Rice Research Institute (IRRI) and the International Maize and Wheat Improvement Center (CIMMYT), while acknowledging the practical challenges posed by time zone differences for sustained coordination.

8.3 Industry engagement and data sharing

Participants discussed international partnerships involving data sharing and industry collaboration, including potential partnerships with seed companies, livestock organizations, and technology providers, while emphasizing the need for standards to integrate different data types. The benefits and challenges of collaborating with industry were debated, with participants arguing for moving forward with partnerships despite data security concerns and for outreach to instrument companies to make data sharing easier. Additional opportunities were identified with germplasm centers and organizations such as the U.S. Department of Agriculture (USDA) and the French National Research Institute for Agriculture, Food and Environment (INRAE), with an emphasis on developing standards and making data useful for both the public and private sectors.

8.4 Funding mechanisms and strategic alliances

The session identified concrete mechanisms for international collaboration and resource sharing. These included potential partnerships with Australia's Commonwealth Scientific and Industrial Research Organisation (CSIRO); participation in European Cooperation in Science and Technology (COST) Actions; and joint funding opportunities involving the U.S. National Science Foundation (NSF) and the U.K. Biotechnology and Biological Sciences Research Council (BBSRC), along with funding opportunities involving Ireland, Israel, and the United Kingdom. Specific applications of AgBioData resources were highlighted for hypothesis-driven research and for improving seed quality across different ecological zones. Collaboration with ELIXIR, the European intergovernmental life-science data infrastructure, was noted as a route toward developing standards across resources, with presentations to the established ELIXIR Plants and Domestic Animals communities identified as near-term engagement opportunities. Participants also discussed partnerships with nongovernmental organizations and governments for smaller-scale agricultural projects, particularly around genetics and plant varieties. Based on their complementary roles in data coordination, functional annotation, and genotype-to-phenome infrastructure, the authors identify EMBL's European Bioinformatics Institute (EMBL-EBI), the Functional Annotation of Animal Genomes

(FAANG) consortium, and the Agricultural Genome to Phenome Initiative (AG2PI) as additional prospective partners for further exploration.

9. Data Infrastructure as a Socio-Technical System

The meeting outcomes reinforce the view that agricultural genomics infrastructure is a socio-technical system. Technical components such as APIs, ontologies, metadata schemas, and genome browsers are necessary but insufficient. Their effectiveness relies on governance structures, community incentives, curation labor, contributor recognition, training, and long-term institutional commitments (Harper et al., 2018; Wise et al., 2019; Wilson et al., 2021).

Many barriers to reuse arise not because standards are absent, but because the social systems required to implement them are incomplete. Authors may lack incentives or guidance to submit reusable data. Journals may not have workflows for validating domain-specific metadata. Databases may lack funding for curation. Curators may lack formal credit for the intellectual labor involved in data harmonization. Users may not know which repositories, ontologies, or formats are appropriate (Harper et al., 2018; Clarke et al., 2023; Hafner et al., 2025). These social and organizational gaps directly affect technical outcomes.

These technical and social mechanisms are mutually dependent. Standards without guidance, incentives, and trained maintainers may not be adopted; repositories without stable governance and funding cannot guarantee reliable access; and attribution mechanisms cannot produce meaningful credit unless journals, funders, and research communities recognize data-stewardship contributions.

Recognizing the socio-technical nature of data infrastructure reframes the solution space. Improving agricultural genomics data reuse requires not only better tools but also clearer expectations, shared governance, workforce development, credit mechanisms, and sustainable funding (Wise et al., 2019; Wilson et al., 2021; Lin et al., 2024). Investment in these areas reduces the time researchers spend locating, reconciling, and recreating data; enables evidence to be combined across databases, species, and institutions; and accelerates its use in crop and livestock improvement, pest and disease response, and other agricultural applications. Without such investment, valuable data may remain technically available but practically unusable, specialized knowledge may disappear as resources are lost, and public funds may be spent repeatedly reconstructing prior work. This framing is particularly relevant to community-owned and community-led journals because peer review can reinforce community norms around data quality, transparency, and reuse.

10. Lessons Learned and Future Directions

The AgBioData 2026 Annual Meeting underscored that agricultural genomics databases are entering a new phase of development. Earlier community efforts focused on identifying shared challenges and developing recommendations for FAIR data stewardship and database sustainability (Harper et al., 2018; Deng et al., 2023). The 2026 meeting shifted the emphasis toward implementation: embedding better data practices into publication workflows, preparing datasets for computational reuse, coordinating federated infrastructure, and sustaining the people and systems that make agricultural genomics data reusable (Wilkinson et al., 2016; Hafner et al., 2025).

Several lessons emerged. First, FAIR practices are most effective when implemented early, ideally before or during publication, rather than retrofitted after data have already been released (Wilkinson et al., 2016; Andrés-Hernández et al., 2021). Second, AI-readiness should be discussed in relation to FAIR data stewardship without reducing FAIR to AI optimization. Machine-readable formats, interoperable identifiers, rich metadata, provenance, and versioning support both human and computational reuse, but the values guiding data stewardship remain broader than AI utility alone (Leipzig et al., 2021; Rocca-Serra et al., 2023). Third, agricultural genomics infrastructure is unlikely to be solved through centralization. A federated model that supports shared services, common standards, and cross-database interoperability better reflects the diversity of crops, organisms, communities, and data types represented across AgBioData resources (Harper et al., 2018; Devare et al., 2023; Deng et al., 2023).

The meeting also reinforced that sustainability is not only a technical problem. Databases depend on curators, developers, community governance, documentation, training, and long-term institutional memory. These functions are often invisible in short-term grant models but are essential for reliable data reuse and continued community value (Harper et al., 2018; Devare et al., 2023; Hafner et al., 2025).

Future efforts should prioritize action across the research lifecycle: data generation, publication, repository deposition, database curation, computational access, governance, and funding. Near-term priorities include piloting FAIR-at-publication workflows, developing practical criteria for AI-ready agricultural genomics datasets, improving dataset attribution and observability, and identifying shared services that can reduce duplication across databases (Leipzig et al., 2021; Lin et al., 2024; Haley et al., 2026). Longer-term priorities include strengthening international coordination, developing sustainable governance models, and advocating for agricultural genomics databases as core research infrastructure rather than short-term research outputs (Harper et al., 2018; Devare et al., 2023).

Taken together, these priorities position AgBioData not merely as a forum for coordination among agricultural databases, but as a community capable of defining, testing, and advocating for the shared practices and infrastructure required for durable agricultural genomics data reuse.

Supplementary Material

Supplementary File 1. Meeting schedule for the 2026 AgBioData Annual Meeting.

Supplementary File 2. Summary of the Working Group sessions at the 2026 AgBioData Annual Meeting

Acknowledgements

The authors thank the AgBioData Consortium members, working group leads, invited speakers, panelists, and meeting participants whose discussions informed this report. This work was supported in part by National Science Foundation Research Coordination Network award 2126334 and by the U.S. Department of Agriculture, Agricultural Research Service. The content is solely the responsibility of the authors and does not necessarily represent the official views of the funders or the authors' institutions. USDA is an equal opportunity provider and employer. During the preparation of this work, the authors used Zoom AI Companion to generate automated summaries as well as transcripts of meeting discussions and ChatGPT 5.6 (OpenAI) to assist with summarizing and synthesizing meeting notes and with manuscript organization, editing, and language refinement. These tools supported, but did not replace, the authors' interpretation of the meeting discussions. The authors have reviewed and revised all AI-assi sted content and take full responsibility for the content of the published article.

Conflict of interest statement

The authors declare no competing financial or personal interests related to this manuscript.

Ethics statement

This meeting report summarizes community discussions and does not report human subjects research, animal research, or experimental work requiring ethics committee approval.

Data availability statement

No new research datasets were generated or analyzed for this meeting report. Informal participant polls were used to structure meeting discussion and are summarized narratively rather than treated as research datasets. The meeting schedule is included as Supplementary File 1, and summaries of the working-group sessions are included as Supplementary File 2.

Originality and copyright statement

The authors declare that the submitted work is original, has not been previously published, is not under consideration elsewhere, and does not breach copyright.

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Supplemental File 1

Supplemental File 1. An Overview of the 2026 AgBioData Annual Meeting Schedule

2026 AgBioData Annual Meeting

Dates: April 20th – April 22nd, 2026
Location: Olathe, KS, USA

Workshop Goals

This workshop is designed to be more than a check-in. It's a moment to take stock, make decisions, and leave with a clear path forward. Over the course of our time together, we aim to celebrate and demonstrate what AgBioData has accomplished under the current grant, while defining the concrete next steps that will carry the network forward. Sustainability sits at the heart of our agenda, both at the network level and for individual databases. We'll also revisit the core RCN objectives (community building, broadening participation, education, and sustainability) to assess where we've made progress and where work remains. By the time we wrap up, the goal is simple: real actions, clear owners, and genuine commitments.

Workshop Structure: Four Outcome-Focused Half Days

Half Day	Focus
Mon PM	Dedicated to demonstrating the group accomplishments under the current RCN and setting the stage for defining our future goals.
Tue AM	Finding our *what* and defining our *when*. We will reflect international priorities and progress with our working groups to clearly define network future goals.
Tue PM	A deep dive into strategies for database sustainability.
Wed AM	Discussing our *how*. A broader discussion on network sustainability, before we close with clear action items for the future.

Monday, April 20th, 2026 (First half day)	
11:00 AM – 11:15 AM	**Welcome & Orientation for Online Participants** **Session Description:** This virtual-only session gives remote attendees a chance to get oriented - how to use Zoom, how to engage in Slack, and how our daily Virtual Watch Parties fit into the experience. We will guide you through it all and stick around for questions, so you can hit the ground running when the workshop kicks off. **Session Material(s):** Pre-recorded video
4:00 PM – 4:10 PM	**Hybrid Welcome & Orientation** **Session Description:** A quick but important 10 minutes to orient everyone before we dive in. We'll cover our overarching goals, the four half-day structure, and how decisions, notes, and action items will be captured throughout the workshop. Hybrid logistics - recording, asynchronous access, and the Slack channel - will get a brief rundown. **Session Material(s):** None
4:10 PM – 5:00 PM	**PI Report: Grant Objectives & Progress** **Session Description:** Where have we been, and where are we headed? This session traces the history of AgBioData and the Research Coordination Network (RCN), celebrating key accomplishments while taking an honest look at the road ahead. As current funding draws to a close, this overview frames the central question driving our workshop: how do we ensure the long-term sustainability of the network and its databases? **Session Material(s):** None
5:15 PM – 5:45 PM	**Grant Writing Team Report** **Session Description:** Which funding opportunities are we chasing, and why? This session summarizes the grants our Grant Writing team has evaluated and selected to pursue, walking through timelines and the resources required. It also serves as a preview of our final day, when the full group will come together to dig into the needs, challenges, and leadership - both collective and institutional - that will shape our path forward.

	Session Material(s): None

Tuesday, April 21st, 2026 (Second half day)	
9:00 AM – 10:00 AM	**Discussion: International Priorities for the Next Phase of the AgBioData Consortium** **Session Description:** Where does the AgBioData Consortium fit in on an international scale? Our international participants bring valuable perspectives that we are committed to better integrating into our work. Expanding partnerships with international databases and consortia that have complementary missions presents both a strategic priority and a natural next step. As AgBioData looks ahead, this session seeks to answer a key question: how do we position ourselves as a genuinely global consortium that not only welcomes international partners, but actively benefits from and contributes to the broader agricultural ecosystem? **Session Material(s):** None
10:00 AM – 11:00 AM	**Working Group Brief Introductions & Reports** **Session Description:** Our four active and two recently completed working groups will introduce their work to the AgBioData community and set the stage for breakout discussions in the next session. ● Standards for Genetic Variation Data ● Phenotypic Data Standardization and Management ● AI-Readiness ● FAIR Scientific Literature ● Natural Language Processing for Biocuration ● scRNA-Seq Biocuration
11:15 AM – 12:45 PM	**Working Group Sessions** **Session Description:** A focused opportunity for four of our active Working Groups (AI-readiness, FAIR Scientific Literature, NLP for Biocuration, Single Cell RNA-Seq) to engage with the consortium on their progress and goals. Each moderated, 45-minute session will provide feedback that will help drive each working group forward. **Session Material(s):** None

1:45 PM – 2:45 PM	**Working Group Report-Backs** **Session Description:** Designated speakers from the previous session's breakout rooms will share feedback, insights and next steps for each working group. **Session Material(s):** None

Tuesday, April 21st, 2026 (Third half-day)	
3:00 PM – 4:00 PM	**Database Sustainability Deep Dive: Presentation and Panel** **Session Description:** The session will be split into a presentation from the Sustainability Working Group, followed by a panel discussion on sustainability of Genomic, Genetic, and Breeding (GGB) databases and digital resources. **Session Material(s):** None
4:30 PM – 5:45 PM	**Database Sustainability Working Session: 1- & 2-Pagers** **Session Description:** This session is dedicated to discussing how to increase database visibility using the popular 1-pager and 2-pager formats. Join us to actively work on creating a communication piece for the AgBioData network as well as for your database to stakeholders and funders across scientific, industry, legislative, and general audiences. We'll provide a starting template, and work with you to tailor it for each audience. **Session Material(s):** 1-pagers for Scientific/Research, Industry-related and Legislative Audiences, and 2-pagers for individual databases
7:15 PM – 8:00 PM	**Invited Presentation from ID3A** **Session Description:** How do we tackle database sustainability in the age of AI? Join us to hear from Kansas State University's Institute for Digital Agriculture and Advanced Analytics (ID3A) on their mission, collective efforts, and institutional sustainability practices in the face of rapidly changing priorities. **Session Material(s):** None

Wednesday, Apr 22, 2026 (Fourth half-day)	
9:00 AM – 9:30 AM	**Awards Reception** **Session Description:** We will recognize significant efforts from our community members to drive the AgBioData vision forward. This includes community awards for Best Workshop Contributor, Best Working Group Contributor, Mission and Leadership, and FAIRification, as well as Ambassador and Champion awards. **Session Material(s):** None
9:30 AM – 11:00 AM	**Network Sustainability Working Session** **Session Description:** What does a sustainable, independent AgBioData actually look like – and how do we get there? Led by the Grant Writing Team, this session moves from vision to strategy. Marcela opens with a 10-minute focused on funding opportunities, working group priorities, and broader impacts before the group tackles the real work: identifying funding opportunities, recruiting PI leadership at the institutional level, and confronting the thorniest questions around governance, equity across databases, nonprofit status, and independence from any single controlling entity. Through facilitated small group discussions, we'll work toward answers - not just conversations. **Session Material(s):** None

11:30 AM – 12:30 PM	**Network Sustainability Working Session - Commitments & Next Steps** **Session Description:** This session recap of everything discussed in the working session. We'll also wrap with an Open Community session to share challenges, wins, and next steps for network sustainability. By the end of the session, we'll walk away with tangible near-term actions, long-term goals, and specific people committed to following up. **Session Material(s):** None

Supplemental File 2

Supplemental File 2. Summary of the Working Group sessions at the 2026 AgBioData Annual Meeting

AI-Readiness Working Group

The overarching goal of the AIR WG is to explore how agricultural data can be organized, formatted, annotated, and packaged to maximize utility for training, validation, and application of AI models. The AIR WG hosted a hybrid meeting at the Annual Meeting to discuss critical issues in AI-readiness, with participants including members and non-members of the WG. There were four prompts which drove the working group's discussion:

1.) Who is using your data for AI purposes?,

2.) What data, metadata & format(s) do they need?,

3.) What are [the] barriers to producing AI-ready data?, and

4.) What would you like to see as an outcome from the WG?

A recurring theme was the visibility gap in database users. For many of the databases, there is little visibility into whether the users behind their daily traffic are researchers, students, or automated AI agents. Without understanding which data is being accessed and by whom, it becomes difficult to identify concrete directions for AI-readiness (i.e., which data/metadata are needed, and what are the applicable barriers).

A significant portion of the discussion thus addressed the practical challenges of managing AI-driven bot traffic, which remains one of the biggest hurdles in identifying legitimate database users. Multiple members commented that bot traffic from supercomputing centers is already causing access issues, and causing some institutions to risk being blacklisted from vital resources. One approach that has shown promise is requiring registration and tokens for API access, which gives resource providers better visibility into usage patterns and some control over misuse, though it introduces its own burdens around security, cost, and personally identifying information.

An overarching question posed to the group was: *what do databases actually need to do differently now, compared to before the "AI boom"*? The group noted that AI users aren't necessarily asking for fundamentally different data than human researchers, but that the data needs to be more consistent, more accessible, and better structured for scale-able ingestion. These actions are particularly important as researchers default to whichever resource has the most interoperable format, even if a more specialized one exists. One example given was how the computational protein structure database AlphaFold (http://alphafold.ebi.ac.uk) uses protein sequence annotations provided by UniProt (http://uniprot.org) even when they do not align with the most recent or

reference annotations used by the species' research community or the species-relevant database. Standardized formats across resources, gold-standard benchmarking datasets, and offering callable query tools to AI agents (e.g., MCP; Model Context Protocol) emerged as promising avenues to better position databases.

FAIR Scientific Literature

The FAIR Scientific Literature Working Group explores barriers and advances in the adoption of FAIR (Findable, Accessible, Interoperable, and Reusable) data practices at agricultural biological databases. There were four prompts which drove the working group's discussion:

1.) What are the incentives for publishers/journals to collaborate with GGB databases?
2.) (How) can we quantify and convey the impact of curated data to publishers?
3.) What critical issues remain for enabling FAIR from start-to-end of publishing models?
4.) What are the barriers to enabling FAIR from the start and how can we reduce them?

To ground the discussion, participants discussed the critical role that publishers and journals play in advancing FAIR data practices and explored opportunities for collaboration between scholarly publishers, community databases, and repositories. Participants noted that while there are existing publisher policies and agreements related to FAIR data, the outcomes of these policies, including how effectively these policies are implemented at the journal and editorial levels, remain unclear. Once more, there was consensus that publishers already understand the *value* of FAIR data, but participants questioned whether publishers currently recognize the benefits of linking publications to curated databases and reusable datasets. Thus a recurring theme was the need to better articulate the value proposition of agricultural biological databases to publishers and journals.

Databases are a source of species-specific genomics expertise and high-quality curated data. From the group members' experiences however, agricultural biological databases are typically excluded from journals' discussions on policies to promote FAIR data. There were several ideas to increase database visibility and collaboration with journals, such as forming relationships with scientists and advisors who work as editors to become advocates of databases, and engaging with journals at meetings and conferences. The group also identified publisher organizations, including STM member journals of the International Association of Scientific, Technical & Medical Publishers

(STM), as potential partners for promoting FAIR data requirements and improving integration between publications and curated data resources. This organization includes journals that publish ~60% of all English language science publications (Science, Nature, Elsevier, OUP, etc).

Several other barriers to FAIR implementation were identified. These included inconsistent enforcement of data availability policies, inadequate metadata, unavailable datasets, variation in editorial oversight, and challenges associated with proprietary or industry-owned data. Participants also highlighted concerns regarding predatory journals and the lack of standardized mechanisms for assessing data quality and FAIR compliance. The discussion underscored the need to clearly define the most significant obstacles faced by publishers, editors, authors, and database curators in order to develop effective solutions.

Education and training emerged as essential components for enabling FAIR data practices from the beginning of the research lifecycle. Participants advocated for training programs that help researchers understand FAIR principles and data management requirements before manuscript submission. Proposed approaches included online learning modules, certification programs, structured publication templates, and standardized reporting frameworks tailored to specific data types. Such resources could improve consistency, reduce barriers to compliance, and facilitate data reuse across research communities.

The working group concluded that advancing FAIR data in scholarly publishing will require coordinated efforts among publishers, journals, repositories, databases, and researchers. Participants identified several priorities for future work, including conducting a gap-analysis of publisher policies and practices, developing FAIRness assessment frameworks, increasing community awareness, evaluating tools that support dataset quality and reuse, and creating mechanisms that reduce the burden of data submission while improving the accessibility and interoperability of research outputs.

Natural Language Processing for Biocuration

The working group examined how artificial intelligence (AI), natural language processing (NLP), and large language models (LLMs) could transform biological database curation and knowledge discovery. There were two prompts which drove the working group's discussion:

1.) If you had one dedicated developer for 6 months to build or integrate an AI tool for your curation workflow, what would you have them work on?

2.) If we wrote a grant proposal to fund shared AI/NLP infrastructure for agricultural databases, what capabilities would need to be in it for you to want your database included?

3.) How should curated databases demonstrate their value in a world where someone can ask an LLM a question and get a pretty good answer? What roles are only possible with curated resources?

Framing the opening question around a fixed resource (one developer, six months) revealed immediate, concrete pain points shared by the databases. Literature mining, document summarization, and knowledge extraction, including systems that allow curators to rapidly trace AI-generated summaries back to supporting evidence within publications, emerged as priority applications. Potential applications included automated identification of gene functions, finding trait-associated accessions, cross-species functional inference, and synthesis of information from large and diverse scientific literature collections. Several databases were already experimenting with such applications, using NLP-based tools like LLM Wiki and ChatPDF for curation and to rapidly check curation sources within a manuscript. Participants noted that AI tools should augment rather than replace human curators, enabling experts to review, validate, and contextualize machine-generated outputs while maintaining scientific rigor.

A recurring theme was the need to balance technological capability with practical constraints. Participants highlighted challenges associated with computational costs, model licensing, access to full-text publications, scalability across scientific domains, and the sustainability of maintaining specialized AI systems. While open-source models offer flexibility, concerns were raised regarding performance, maintenance, and the resources required for continual model training and refinement. However, higher-end models and platforms would require monetary inputs. The group emphasized that future AI infrastructure should support the diverse needs of agricultural databases while remaining transparent, accessible, and cost-effective.

Discussion of the role of curated databases in an AI-driven research environment focused heavily on issues of trust, provenance, and accountability. Participants argued that curated databases provide capabilities that AI systems alone cannot reliably deliver, including expert quality control, validation of conflicting information, standardized metadata, transparent evidence trails, and governance structures that ensure accountability. As AI-generated answers become increasingly common, participants stressed the importance of maintaining authoritative resources that allow users to verify the origin and reliability of information. Concerns were raised that newer generations of researchers may rely on AI-generated responses without understanding the underlying evidence, highlighting the need for education on evaluating scientific sources and distinguishing between generated content and validated knowledge.

The group concluded that curated databases will remain essential components of the scientific ecosystem by serving as trusted, authoritative sources of data and metadata that both researchers and AI systems depend upon. Participants emphasized the importance of developing clear licensing frameworks, tracking AI use of database resources, establishing mechanisms for attribution and accountability, and ensuring that curated resources receive appropriate recognition for the value they provide. The discussion ultimately reinforced the view that the future relationship between AI and biological databases should be collaborative, with curated resources providing the trusted foundation upon which increasingly sophisticated AI tools can operate.

Single-Cell Working Group

The AgBioData Single Cell Working Group focused on four interconnected themes: metadata standards and data reuse, cell-type annotation, pipeline benchmarking, and community infrastructure for agricultural single-cell RNA sequencing (scRNA-seq) data. The working group discussion was driven by four questions:

1.) What is the single biggest metadata or data-reuse bottleneck you've personally experienced with agricultural scRNA-seq data, and did you find a workaround?

2.) What analytical tools or pipelines have you adapted for plant or animal single-cell annotation?

3.) What tools or pipeline components should be prioritized for community benchmarking, and what datasets or metrics could serve as cross-kingdom standards?

4.) Across our four priority areas – metadata standards, cell-type annotation, pipeline benchmarking, and training – which activities can we accomplish now without significant funding, and which require coordinated grant support?

Participants identified the lack of standardized protocols for cell and nuclei isolation, incomplete reporting of analysis pipelines, and limited data-sharing infrastructure as major barriers to data reuse. The discussion emphasized the need for a wide-reaching community initiative to establish and publish metadata standards for agricultural scRNA-seq data, and collaboration with journals to enforce the standards. Participants also stressed the importance of harmonizing vocabularies and annotation standards across datasets and species.

Benchmarking was identified as a high-priority community activity. Participants noted that most benchmarking efforts and analytical tools have been developed around human and mouse datasets/systems, creating an urgent need for agricultural model systems and species-specific workflows. There is no compendium or public catalog on

pipelines used for agricultural scRNA-seq data analysis, and participants emphasized the current lack of well-curated, species-specific annotation resources for scRNA-seq data, further complicating benchmarking efforts. Thus participants proposed creating a public catalog of pipelines used in published agricultural scRNA-seq studies and establishing shared datasets with known or matched “truth” samples for evaluating deconvolution and annotation methods. Reproducible workflows based on frameworks such as Nextflow and Snakemake were viewed as particularly important, and the group discussed developing scoring systems to compare tool performance across plant and animal datasets. Existing efforts, including pipeline comparison tools such as CATD, were highlighted as valuable starting points for collaborative benchmarking initiatives.

The working group also considered what could be accomplished immediately versus what would require coordinated funding. Participants agreed that benchmarking exercises, training activities, outreach programs, and community repositories for tools and annotations could begin with relatively modest resources. More ambitious goals, such as automated ingestion of plant datasets into major repositories, development of cross-species reference atlases, and large-scale comparative analyses, would likely require dedicated grant support and coordination with initiatives such as the Plant Cell Atlas and the European Bioinformatics Institute (EBI). The group also discussed the potential to develop automated annotation approaches, including AI-assisted generation of cell-type markers.

As a practical path forward, the group proposed a proof-of-principle effort using a limited set of agricultural single-cell datasets that could be deposited into public repositories and reused for marker identification, deconvolution, and comparative analyses. Initial species priorities included sorghum, rice, maize, pig, chicken, and cattle, with the expectation that successful workflows could later be adapted across the broader agricultural research community.

Cover Letter

Cover Letter

Dear Editors of Genetic Resources,

We are pleased to submit “AgBioData Annual Meeting 2026: Toward Sustainable Agricultural Genomics Infrastructure” for consideration in the Other papers section of Genetic Resources. This meeting report synthesizes the outcomes of the 2026 AgBioData Annual Meeting and places them in the context of the consortium’s sustained work to improve reuse, interoperability, and sustainability across agricultural genetics, genomics, and breeding databases.

In the article, we argue that agricultural research increasingly depends on distributed digital resources connecting genomic variation, phenotypes, germplasm, environmental observations, and the scientific literature. Yet these resources frequently rely on inconsistent metadata, identifiers, technical standards, and funding models. Data may therefore be publicly available while remaining difficult to combine or reuse across databases, species, institutions, and countries. The manuscript moves beyond identifying these problems to articulate a practical implementation agenda: embedding findable, accessible, interoperable, and reusable practices in publication workflows; preparing data for responsible computational and artificial-intelligence-enabled reuse; developing federated services; and sustaining the people and institutions that make the infrastructure work. These activities are inseparable from the long-term sustainability of breeding, genetics, and genomics databases and resources.

The meeting also marked an important moment in AgBioData’s development. Although the consortium has been organized primarily through a U.S.-based National Science Foundation Research Coordination Network, it has always included international participants and relationships. The 2026 meeting made clear that AgBioData’s future depends on becoming more deliberately international in both participation and partnership. Agricultural genetics and genomics are inherently global, and the consortium must better reflect the worldwide distribution of expertise, data resources, crops, livestock systems, and research priorities. Participants therefore treated international coordination not as a secondary outreach activity, but as an infrastructure and sustainability requirement. The report identifies opportunities to broaden participation, engage underrepresented regions, strengthen relationships with international organizations and database communities, and build standards and shared services that can operate across national and institutional boundaries. With its international readership and focus on the documentation, conservation, management, and use of genetic resources, Genetic Resources is a particularly appropriate venue for connecting these communities at this important moment in AgBioData’s development.

The manuscript fits the journal's scope because it addresses the documentation, management, accessibility, and long-term use of genetic resources through infrastructure, standards, governance, and community coordination. Although the report does not present a conventional experimental study, it provides actionable recommendations for journals, repositories, databases, funders, and research communities whose decisions determine whether genetic-resource data remain usable.

The manuscript is original, has not been previously published, and is not under consideration elsewhere. No new research datasets were generated or analyzed. The meeting schedule and working-group summaries are provided as two supplemental files, and all applicable conflict-of-interest, ethics, data-availability, originality, funding, and artificial-intelligence-use statements are included in the manuscript.

Thank you for considering this manuscript. We believe it will be of interest to readers working across genetic-resource documentation, data stewardship, database development, research infrastructure, and international coordination.

Sincerely,

David C. Molik, Ph.D.

Associate Professor and Department Head

Open Publishing Exchange, Kansas State University Libraries

Member, AgBioData Steering Committee, Co-PI AgBioData

On behalf of the authors